

Agentic Commerce: A Comprehensive Analysis of Cybersecurity Risks, Privacy Challenges, and Trust Mechanisms in Autonomous AI-Driven Marketplaces

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Abstract—The evolution of agentic commerce shows a paradigm shift in digital marketplaces, where autonomous AI agents perform transactions, provide personalized experiences, and make purchases with minimal human intervention. This research paper presents an extensive analysis of the cybersecurity and privacy risks along with comprehending the trust mechanisms implicit in agentic AI powered commerce systems. In this paper, we perform a detailed analysis of current market trends, security vulnerabilities, and case studies of the agentic commerce implementations. We also discuss the threat vectors associated with prompt injection, data poisoning, multi-agent cascading failures, and identity management vulnerabilities.

Index Terms—Agentic AI Commerce, Cybersecurity Risks, Regulatory Compliance, Privacy-Preserving AI, Federated Learning

I. INTRODUCTION

Digital commerce ecosystem is going through a fundamental transformation with the emergence of Agentic AI powered commerce where the agents act as a medium between paying consumers and sellers/merchants. According to recent market data, the north American retail providers saw a 4,700% jump in their website traffic mainly due to the AI agents/browsers crawling the sites for product information [2]. Another interesting data point is that these AI agents spend 32% more time on the site, and the global AI agentic market is expected to grow from a mere 7.55 billion USD in 2025 to a whopping 199 billion USD by 2034. If we calculate the compounded annual growth rate (CAGR) this stands at 43.84% [3].

In this paper, we analyze and discuss the cybersecurity risks and challenges in maintaining privacy guidelines due to the use of Agentic AI commerce pipelines. We also discuss the recent technological challenges and some real-world case studies to understand the true impact of these agentic commerce platforms.

The paper is organized into the following 7 sections. Section II is about the current market landscape and adoption trends with respect to AI agents in commerce, section III will provide a comprehensive threat analysis on cybersecurity risks, section IV will focus on regulatory and privacy challenges, section

V will delve into managing zero-trust in agentic commerce, section VI will list the advanced security mechanism and finally section VIII will list out the recommendations and the conclusion of this research.

II. MARKET LANDSCAPE AND ADOPTION TRENDS

The Agentic AI commerce platform has seen an explosive growth in several dimensions across capitalizing the market, enterprise level adoption and engaging consumers.

A. Market Size and Growth Projections

Research report from Precedence shows that the AI genetic market had a valuation of 5.25 billion USD back in 2024, but the recent projections show it grew to 7.55 billion USD in 2025. The explosive growth will ultimately expand up to 199.05 billion USD by 2034 (CAGR 43.84%) [3]. Especially in the e-commerce market, if we focus just on the e-commerce market, use of Generative AI is also projected to increase 3.519 billion USD by 2034 from 833 million USD in 2024. This research also shows that North American enterprises and consumers are in favor of adoption since they hold 46% market share in 2024.

TABLE I
MARKET GROWTH PROJECTIONS

Year	Agentic AI (USD B)	GenAI E-com (USD B)	CAGR (%)
2024	5.25	0.833	—
2025	7.55	—	43.84
2030	—	—	—
2032	—	3.519	—
2034	199.05	—	43.84

B. Enterprise Adoption Statistics

A recent enterprise-wide survey [4] from PricewaterhouseCoopers (PwC) shows that agentic AI adoption is fast tracked with roughly 79% of surveyed organization reporting adoption

while 19% of the organizations show maturity at scaling the adoption.

Survey also reveals that 35% of the organizations are utilizing AI agents for quick delivery of proof of concepts (POCs)/pilots and whereas 25% of the companies show that they limit the AI agent usage to select areas for maximum gain.

Among the large enterprises with more than 1000 employees, adoption and deployment of AI agents is around 68%, a little lower compared to smaller enterprises but another survey from Cloudera [5] shows that almost 96% of the leaders across Information Technology organizations are planning to expand the usage and adoption of Agentic AI in the near future.

TABLE II
ENTERPRISE ADOPTION STATISTICS

Organization Type	Adoption (%)	Status
All Organizations	79	19% scale, 35% pilots
Enterprises (1000+)	68	68% deployed
Fortune 500	79	Active projects
With GenAI	25	Deploying 2025
Planning Expansion	96	Next 12 months

C. Traffic and Engagement Metrics

A recent article from Adobe Analytics shows that the traffic generated by agentic browsers and AI agents increased the overall site traffic by 4700% across major U.S retail providers [2] showing the shifting landscape from user-focused e-commerce platforms to agentic commerce marketplaces.

These AI powered searches and site visits saw an increase of time spent by 32% and page views by 10% across the sites compared to traditional user-initiated sessions. This trend also has unintended consequences as the bounce rate of searches dropped by 27% as the AI agents can make a purchase with lower bias. Boston Consulting Group reports [1] that half of the users across the platforms are leaning towards AI adoption for ease of purchasing and are expected to use some form of agentic commerce by end of 2025.

TABLE III
TRAFFIC AND ENGAGEMENT METRICS

Metric	Value	Period
AI browser traffic YoY	+4700%	July 2025 vs 2024
Time on site	+32%	vs traditional
Pages viewed	+10%	vs traditional
Bounce rate	-27%	vs traditional
Consumers using AI	50%+	End of 2025

III. CYBERSECURITY THREAT LANDSCAPE

A. OWASP Agentic AI Threat Taxonomy

The OWASP Agentic Security Initiative explores the emerging security implications owing to the adoption of AI systems utilizing advanced agentic AI frameworks [7]. The initiative has classified the threats into 15 distinct categories mapping them to the architectural components of the agents.

According to OWASP, Agentic AI threats are either new or agentic variations of existing threats. Some notable threats are the result of new components agentic AI application architecture brings.

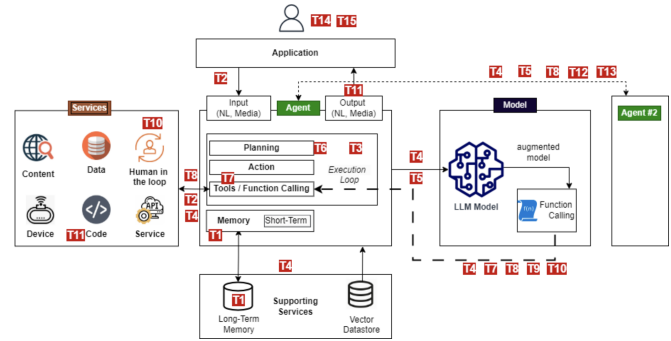


Fig. 1. OWASP Agentic AI Threat Model Summary

TABLE IV
OWASP THREAT TAXONOMY

ID	Threat	Impact
T1	Memory Poisoning	High
T2	Tool Misuse	Critical
T3	Privilege Compromise	High
T4	Resource Overload	Medium
T5	Cascading Hallucination	High
T6	Intent Breaking	Critical
T7	Deceptive Behaviors	High
T8	Untraceability	Medium
T9	Identity Spoofing	High
T10	Overwhelming Human	Medium
T11	RCE/Code Attacks	Critical
T12	Comm. Poisoning	High
T13	Rogue Agents	Critical
T14	Human Attacks	High
T15	Human Manipulation	Medium

B. Documented Vulnerabilities

According to OWASP Top 10 for LLM applications, Prompt injection is the #1 ranked vulnerability, affecting almost 73% of current deployments. An example of the impact of this vulnerability is evident from the recent data leak of sensitive information from a major financial services firm often featuring in Fortune 500 [8] [9].

Industry research demonstrates that LLMs and AI agents are getting embedded deep into everything from customer interactions to internal company operations, while developers and security professionals are discovering new vulnerabilities and identifying ways to counter them effectively.

Organizations and end users should be aware of how to safely interact with LLMs/AI Agents. They need to understand the risks of unintentionally providing sensitive data, which may later be disclosed with as a model output of agentic action. A recent analysis on identity risks [11] reveals that some unintended security consequences in nearly 80% of organizations utilizing AI agentic capabilities such as com-

promising long lived access tokens used by Agentic AI SDKs and browsers.

TABLE V
MAJOR SECURITY VULNERABILITIES

Vulnerability	Prevalence	Mitigation
Prompt Injection	73% deployments	High
Data Poisoning	1% samples	Very High
Tool Manipulation	80% unintended	Medium
Credential Exposure	100% of SDKs	Medium
Agent Hijacking	Major platforms	High
Cascading Failures	Emergent	Very High

IV. PRIVACY AND REGULATORY COMPLIANCE

A. Consumer Privacy Concerns

Recent consumer surveys reveal several privacy and technical concerns related to the use of AI agent powered commerce platforms and websites. One of the most crucial concerns stems from securing payment information of the users as nearly 32% of the enterprises identify that they are worried about unauthorized transactions and fraud if the AI agents are supplied with access and autonomy to make purchases using users' payment credentials. This is evident from some of the advancements we recently witnessed in protecting agentic payment solutions.

While privacy is often dictated by several factors, nearly 26% of the enterprises are concerned about the breach of regulatory requirements such as GDPR/CCPA intentionally/unintentionally.

The survey also shows some minor concerns (18%) about incomplete/incorrect purchases made by these agents resulting in affecting the supply chain and inventory of the products and returns while some enterprises roughly 17% fear that the autonomous decision-making power of the agents may result in loss of control when it comes to user needs.

This rapid adoption also brings in new challenges to the fundamental foundation of these e-commerce sites. As evident from the recent announcement from Amazon that they are suspending Perplexity.AI's shopping agent, Comet from making purchases on its platform. Amazon claimed the AI agent violates its terms by not identifying itself as a bot and engaged in automated purchasing which is against Amazon's policies [42].

This tension between convenience and privacy presents a unique but critical for agentic commerce deployment in contemporary industries.

TABLE VI
CONSUMER PRIVACY CONCERNS

Concern	Consumers (%)	Regulatory Framework
Payment Security	32	PCI-DSS, ACP, AP2
Privacy/Data Protection	26	GDPR Art 22, CCPA
AI Mistakes	18	Right to contest
Loss of Control	17	Transparency
Trust in AI	36	Privacy by Design

B. GDPR Compliance Challenges

Adhering to European Unions' General Data Protection Regulation (GDPR) regulation [14] has presented some challenges to the enterprises focusing on the implementation of agentic AI systems especially in the consumer-focused commerce industry.

Some of the main challenges include but are not limited to GDPR's Article 6 and Article 22 requirements about consent before processing user's data and human oversight becomes problematic to the AI commerce pipelines as the agents are constantly learning and adopting often autonomous decisions which might sometimes overlook the user consent in processing their sensitive data.

Also, Articles 13 and 14 require organizations to be transparent in storing and processing personal data of users, which becomes complicated as the AI agents may process data with less transparency, and the complexities of the data processing may not be immediately clear or easy to understand by the users.

C. Emerging Protocols

There is also positive side to the privacy and security concerns which often lead to new innovations in the field as seen in the recent press release by the world's leading payment solution provider Visa announcing a new concept called 'Trusted Agent Protocol', establishing a foundational framework for agentic commerce that enables secure communication between AI agents and merchants during every step of a transaction [41].

Open AI and Stripe partnered to create an Agentic Commerce protocol which enables users of the Open AI ChatGPT platform to purchases within their ecosystem without compromising the payment credentials by leveraging encrypted form of payment tokens [15] [16].

Other mentions include the Agent Payments Protocol (AP2) for establishing standards for agent-to-merchant payment interactions with user presence verification [18], Web Bot Authentication by Cloudflare [17] that enables the commerce platforms to authenticate agents through the use of Trusted Agent Protocol and last but not least the Secure Payment Tokens (SPTs) by Stripe that provides granular controls for identity and trust management.

V. ZERO TRUST ARCHITECTURE

A. Zero Trust Principles

Cybersecurity entered a new phase with threats as the advent of AI powered platforms which not only exploit software but understand and exploit the natural language processing for malicious gain. Agentic commerce fueled by AI agents are not only about running instructions, but they are designed to understand natural language and learn from human behavior to optimize their functionality and delivery.

So, the traditional Zero trust models which usually focuses on "never trust, always verify" for users or the devices within a network doesn't necessarily apply to the agentic AI landscape. In this paper, we focused on the Zero Trust principles that not only verify users/devices but also the identity of agents making

the requests, their intent and whether that intent aligns with the authorization. This shift is needed to look beyond network and identify and focus on behavior and intent for good security measures.

Some of the core Zero Trust principles for securing AI agents architecture include the following.

Mapping, discovering and creating an inventory all AI agents and their identity and access roles, managing appropriate permission sets leveraging least privilege and intent-based access controls, proper governing of asset ownership and enforcing appropriate lifecycle policies, performing continuous authentication/authorization through behavioral biometrics and implementing micro segmentation by performing network isolation and preventing parallel movements within the network with elevated access [23] [24].

B. Security Mitigation Strategies

Common security mitigation strategies may not help with the rapidly evolving agentic commerce landscape.

Focusing on the basic principle of defense in depth and leveraging appropriate governance risk frameworks such as NIST RMF and ISO 42001 at the same time emphasizing managing agentic identities, providing network isolation, implementing agentic authentication pipelines, sensitive data protection via using homomorphic encryption and continuously monitoring the AI agent usage and statistics using a centralized SIEM tool is needed.

TABLE VII
SECURITY MITIGATION STRATEGIES

Layer	Zero Trust	Complexity
Identity & Access	Least privilege	High
Network Security	Micro-segmentation	Medium
Data Protection	Encryption	High
Monitoring	Continuous monitoring	Medium
Governance	Audit trails	Very High
Authentication	MFA, DIDs, VCs	High

VI. PLATFORM IMPLEMENTATIONS

A. Major Platform Deployments

In this research we focused on highlighting the recent implementation of the agentic commerce platforms from leading AI and Fintech companies.

OpenAI has recently launched the 'Instant Checkout' [15], AI agent powered checkout feature within the ChatGPT ecosystem. Using instant checkout users will be able to make purchases on commerce platforms like Etsy and Shopify. This is comprehensive agentic commerce capability as it utilizes Stripe's Agentic Commerce Protocol for payment processing capabilities while natively enabling the checkout experience within the ChatGPT ecosystem.

At the surface it may seem like a common checkout experience similar to a e-commerce site, but OpenAI enabling this to their 20 million strong ChatGPT pro user bases without actually owning the products has serious repercussions to the e-commerce providers and traditional user base.

In addition to the above, Google announced they are introducing Gemini powered AI Mode shopping with Google Pay integration [5] to their user base. Google has been an industry leader in providing personalized experiences to its user base and this new agentic commerce implementation has the potential to reach billions of google users worldwide.

Another important leader in this agentic commerce space is one of the fastest growing companies in the world, Perplexity.AI. They released Perplexity's Buy with Pro in late 2024 [31] with PayPal integration [18].

Similar motions can be said about Shopify's agentic infrastructure using A2A and AP2 protocols along with PayPal's Agentic Toolkit providing AP2-compliant integration and Stripe's Secure Payment Tokens enabling identity management.

TABLE VIII
PLATFORM IMPLEMENTATIONS

Platform	Launch	Key Features
Perplexity	Late 2024	Buy with Pro
ChatGPT/OpenAI	Sept 2025	Instant Checkout, ACP
Google Gemini	2025	AI Mode, tracking
Shopify	2025	Catalog, A2A
PayPal	Oct 2025	Toolkit, AP2
Stripe	Oct 2025	SPTs, ACP

B. Enterprise Case Studies

Enterprise deployments demonstrate measurable impacts as evident from the case studies below.

A leading apparel store company, H&M implemented store layout optimization using AI for optimizing the user conversion rates across its website not just on the physical stores. A company named Zalando introduced dynamic pricing powered by AI which delivered an increase of 12% revenue per SKU and +17% increase in basket size [34].

Amazon/Shopify shopping assistants achieved +25% AOV and -19% return rates [31] [34] by utilizing AI powered commerce engines. SMS and web chat enabled on PacSun's website using AI agents reached 19% conversion with 85% deflection by using existing FAQ.

A delivery automation enterprise ScS saved \$260K annually by enabling agentic AI to optimize their automation pipelines and another agentic powered automation from Easterseals healthcare RCM perlemoen reduced A/R by 35 days and primary denials by 7%.

TABLE IX
CASE STUDY RESULTS

Company	Implementation	Results
H&M	Store layout AI	Daily optimization
Zalando	Dynamic pricing	+12% revenue
Amazon/Shopify	Shopping assistants	+25% AOV
PacSun	Web/SMS agents	19% conversion
ScS	Delivery automation	\$260K savings
Easterseals	Healthcare RCM	-35 days A/R

VII. ADVANCED SECURITY MECHANISMS

A. Federated Learning

As enterprises focus on the adoption of complex commerce landscapes using LLMs and agentic AI systems, to act autonomously on behalf of users, we need a way to effectively preserve privacy and operational security across the ecosystem.

If we leverage the federated learning concepts which shifts from traditional machine learning approaches by embedding personalized learning and providing privacy by design during the training of LLMs, these agentic AI systems can reduce privacy concerns considerably.

Some of the core principles of federated learning such as central training and securing raw data within the source devices will help ensure the agentic AI systems do not have access to the raw data which were used to train the models.

For example, dynamic pricing based on user behavior and fraud patterns can be deducted using AI agents but without compromising the transaction confidentiality. However, it has its own challenges with respect to operational cost of heterogeneous data and communication overhead.

B. Explainable AI (XAI)

If the Agentic Commerce Platform providers invest and implement explainable AI processes that allow human users to understand and accept the results of the agentic decision-making rationale while adhering to regulatory compliance, then it will help secure the overall ecosystem.

This explainable AI implementation can also help in understanding the Bias of the decisions made by the agents. This can resolve doubts about discriminatory practices that may arise from demographic or sociological constraints and could also explain the reason for fraud alerts so users can better leverage the agentic capabilities within their bounds.

C. AI Governance Frameworks

Secure governance of the AI frameworks and agentic AI commerce platforms should not be an afterthought. There are several industry leading frameworks such as NIST AI RMF that emphasize transparency, safety and accountability while the OECD AI principles [26] frameworks prioritize human-centered values and fairness.

IEEE's ethically aligned design framework that shares a set of principles and guidelines for the ethical development of AI and autonomous systems is also a good option in security agentic commerce systems.

TABLE X
AI GOVERNANCE FRAMEWORKS

Framework Name	Organization/Origin
AI Risk Management Framework	NIST
AI Ethics Guidelines	European Commission
AI Principles	OECD
Ethically Aligned Design	IEEE
Secure AI Framework	Google
Responsible AI Standard	Microsoft

VIII. CONCLUSION

In this paper, we performed a comprehensive analysis to show the rapidly evolving agentic commerce ecosystem from the lens of organizational adoption metrics to exponential growth in website traffic, from the ever-changing threat landscape to the regulatory implications and significant security challenges as the security and risk landscape has not kept pace with the innovation and speed of these agentic platforms which focuses on autonomous AI capabilities in commerce operations.

Despite these challenges, we also showed that there are several industry frameworks and protocols such as OWASP Agentic AI security framework, Zero Touch Architecture, Visa's Trusted Agent Protocol and specialized protocols from leading AI and Fintech companies offer mitigation strategies and pathways to secure the agentic commerce platforms.

This rapid adoption and transformation into agentic commerce is inevitable but responsibility can occur by embedding security and privacy into the design of these agentic commerce platforms.

We hope that the frameworks, protocols, and best practices discussed and identified in this research paper pave way for solid building blocks that power the next generation trustworthy agentic commerce ecosystems.

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